

What the Machine Cannot Hear

Evaluating Whisper on South Asian Languages for Diasporic Oral-History Archives

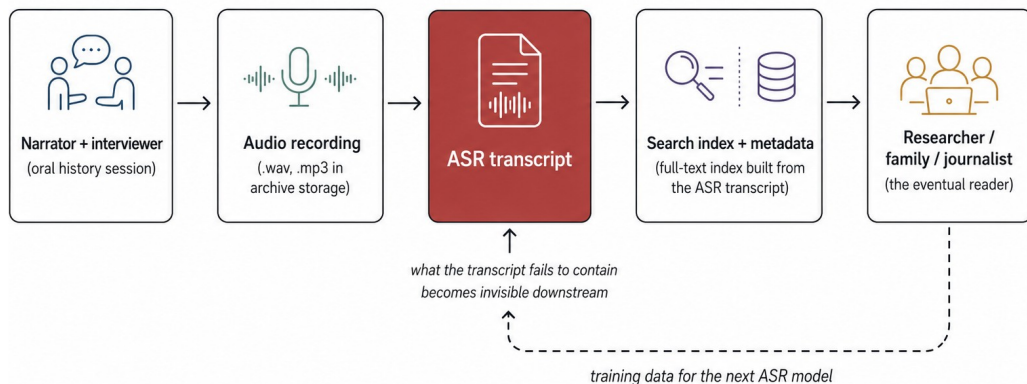
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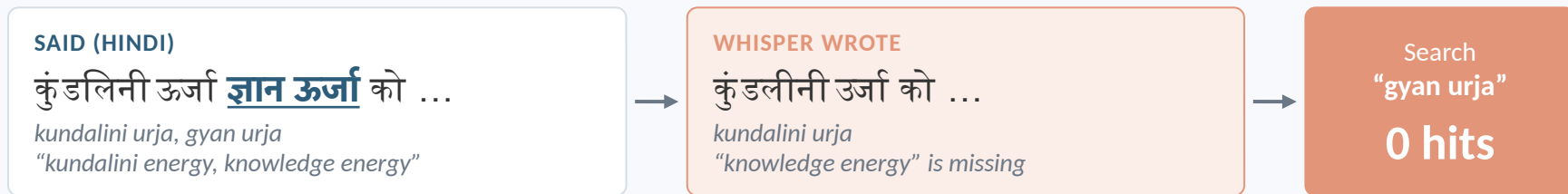


If a word is not in the transcript, it is not findable in the archive

THE DIASPORIC AUDIO ARCHIVE PIPELINE



- **SAADA, 1947 Partition Archive:** thousands of hours of South Asian testimony, often code-switched
- **ASR writes the first transcript.** It drives search, indexing, and future training data
- **Transcripts are interpreted artefacts** [1, 2]: what the model misses, the archive loses



Published WER shows the gap, not what it does to the archive

Known: Whisper WER on South Asian languages is several times English at every model size [3]; off-the-shelf ASR flattens code-mixed speech [4, 5]; ASR disparities are documented across dialects [6].

Missing: which content is dropped, anglicised, or confidently fabricated.

Contribution: a word-level error typology that links each error type to its archive consequence, released as a reviewer checklist.

RQ1 • How large is the gap?

Whisper-large-v3 WER on Hindi, Bengali, Tamil, Urdu vs. English, with bootstrap CIs and corrected tests

RQ2 • Does fine-tuning close it?

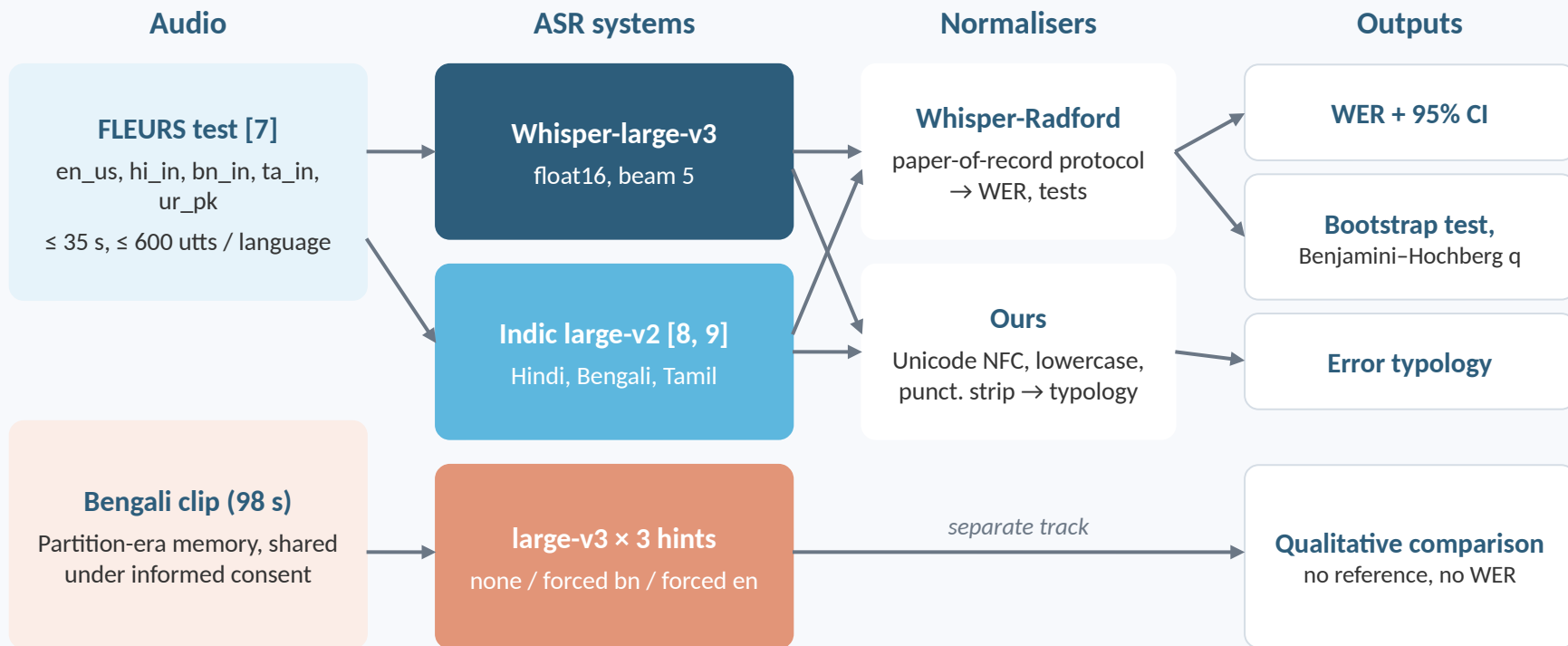
Community Indic Whisper-large-v2 fine-tunes on identical audio, same normaliser

RQ3 • What do the errors look like?

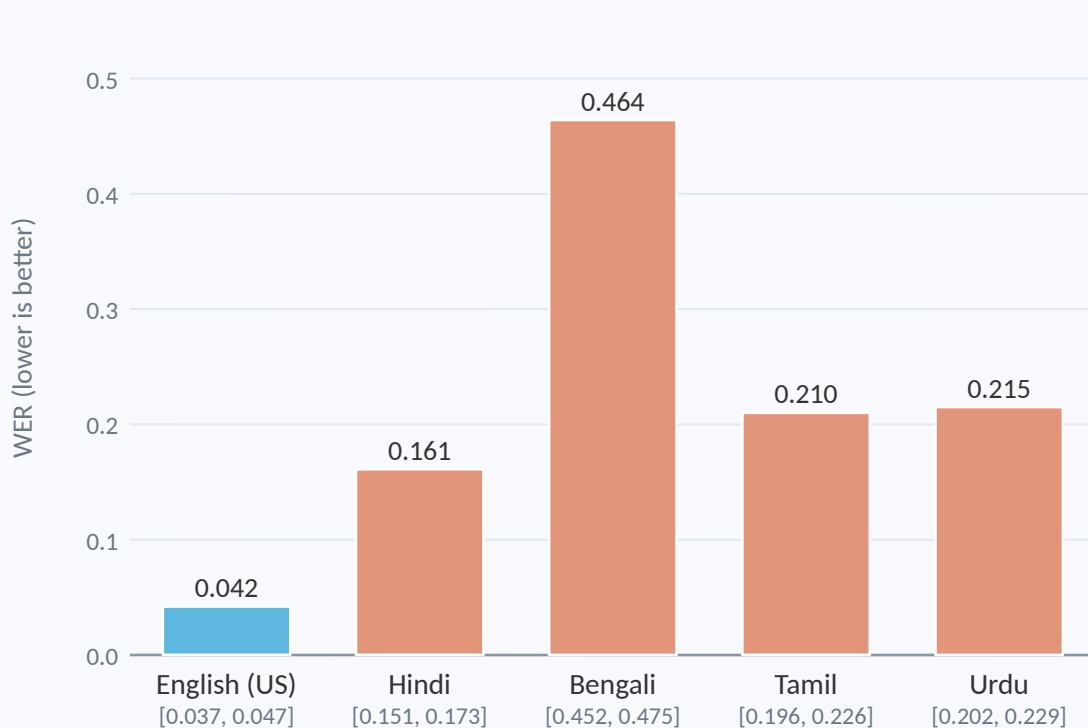
Deterministic word-level typology, plus one real Bengali recording under three language hints



One pipeline: same audio, two classes of model, two normalisers



South Asian WER is 3.8× to 11× the English baseline



Ratio to English

3.8×

Hindi

11.0×

Bengali

5.0×

Tamil

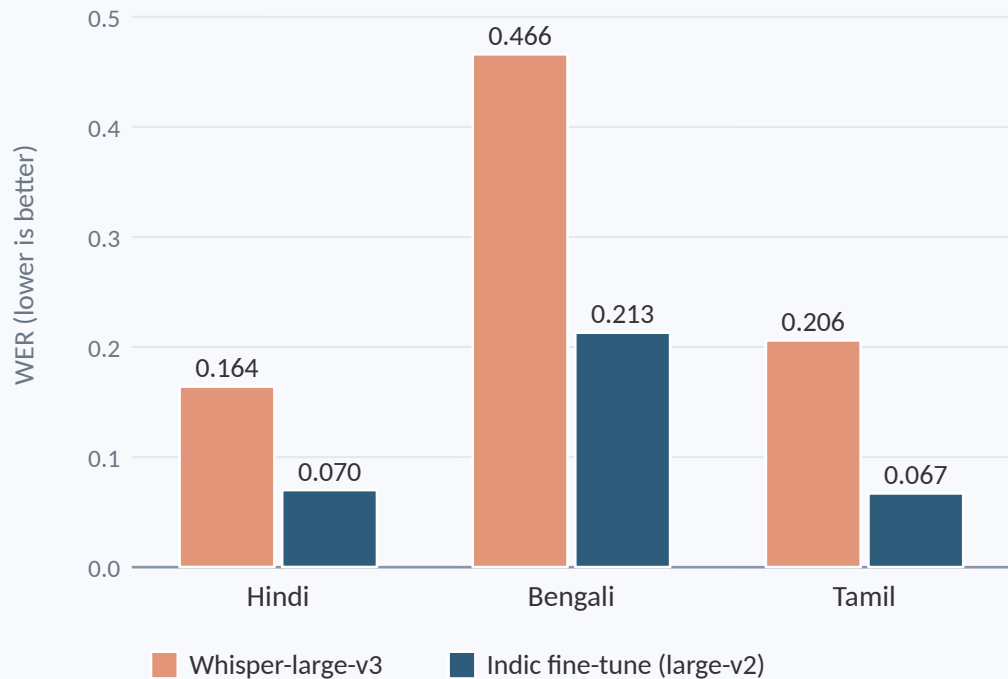
5.1×

Urdu

- **All 4 comparisons significant:**
Benjamini-Hochberg $q \leq 0.001$ (the floor for 1000 resamples)
- **Language identification (LID) ≈ 1.0 :**
Whisper recognised every language correctly; the errors are in transcription



Indic fine-tunes cut WER by 54–67% but do not reach English



After fine-tuning, vs. English (0.042)

Language	WER cut	× English
Hindi	−57%	1.7×
Bengali	−54%	5.1×
Tamil	−67%	1.6×

This sentence: WER 0.10, name lost

SAID डॉ मलार बालासुब्रमण्य म

Dr Malar Balasubramaniam

FINE-TUNE डॉक र मलाल बाला सुब्रमण्य म

doctor malal ("regret") bala subramanyam

Name becomes a common word, surname splits: a search finds nothing.

Upper bounds: Hindi, Tamil fine-tunes saw FLEURS train/dev.



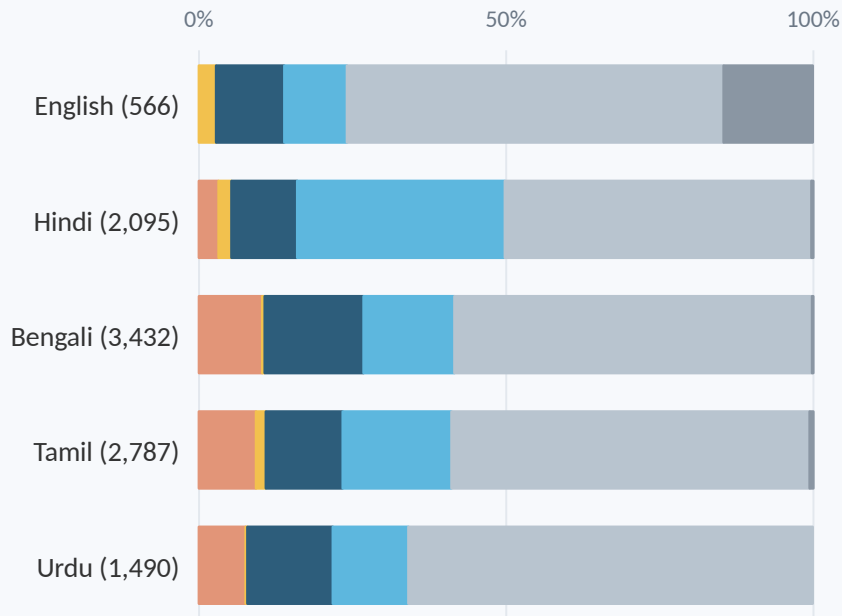
Errors differ in consequence: drops are rare but unrecoverable

Error type	Effect on the archive
Dropped word	Gone from search
Anglicised	Missed in original script
Hallucinated word	False content added
Wrong vowel sign	Near miss
Wrong word	Still searchable
Dropped English word	Gone (mostly English)

Same 600 clips: 360 words tagged as dropped in Bengali, 82 in English. WER weighs a drop like any other error; for an archive it is the costliest.

Error mix per language

Each bar = 100% of that language's errors, not its words. (n) = error count.





Can we trust today's published WERs?

The Whisper paper's per-language tables [3], the Indic fine-tune model cards [9], and our own Table II



No. Published WERs score read speech; archives need conversational ASR

Beyond the caveats already raised (Hindi/Tamil fine-tunes saw FLEURS, the normaliser splits Indic words, one case-study clip), every published WER, ours included, is scored on the wrong kind of audio.

What gets scored: read speech

FLEURS, Common Voice, LibriSpeech-style benchmarks [3, 7, 9]

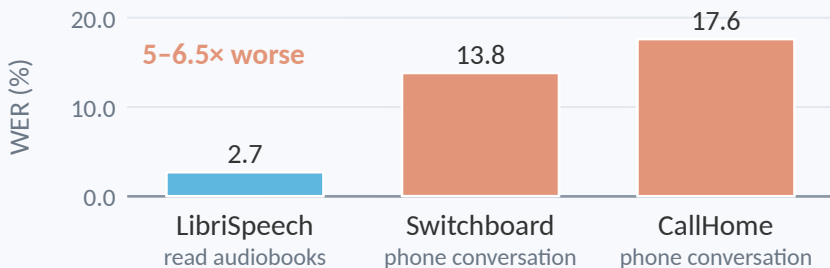
- Sentences read aloud from text by one speaker
- Studio or quiet room, one language per clip
- No hesitation, repair, overlap or code-switch

What an archive records: conversational speech

Oral-history interviews at SAADA, the 1947 Partition Archive

- Unscripted, two voices, pauses, false starts
- Accented diaspora speakers, often elderly, home recordings
- Code-switching mid-sentence; names and places

Same model, same language: English WER, Whisper-large-v2 [3]



Same languages: Whisper WER on read vs spontaneous Indic speech

	FLEURS, read (ours)	IndicVoices, spontaneous [13]
Hindi	0.16	0.34
Bengali	0.46	0.93
Tamil	0.21	0.78
Urdu	0.22	0.30

Our own hint: auto-LID confidence ≥ 0.97 on FLEURS, 0.85 on the spontaneous case-study clip. **Every number today is a floor.**



Forced English: fluent translation and hallucination at LID 1.0

No hint

auto-LID → Bengali, $p = 0.85$

3 Bengali-script segments

Forced Bengali

3 Bengali-script segments

Forced English

20 fluent English segments
reported LID 1.00 (set by the hint)

Why force a language? Pipelines often set it from collection metadata or default to English, so a mislabelled or code-switched recording gets the wrong hint.



Forced-English output (excerpt)

0.5–3.5 s	What is your name?	translation
3.5–5.5 s	[Name]	
5.5–8.5 s	Where were you born? When were you born?	
... 15 more segments ...		
85.8–91.5 s	I died early in my birth.	hallucination

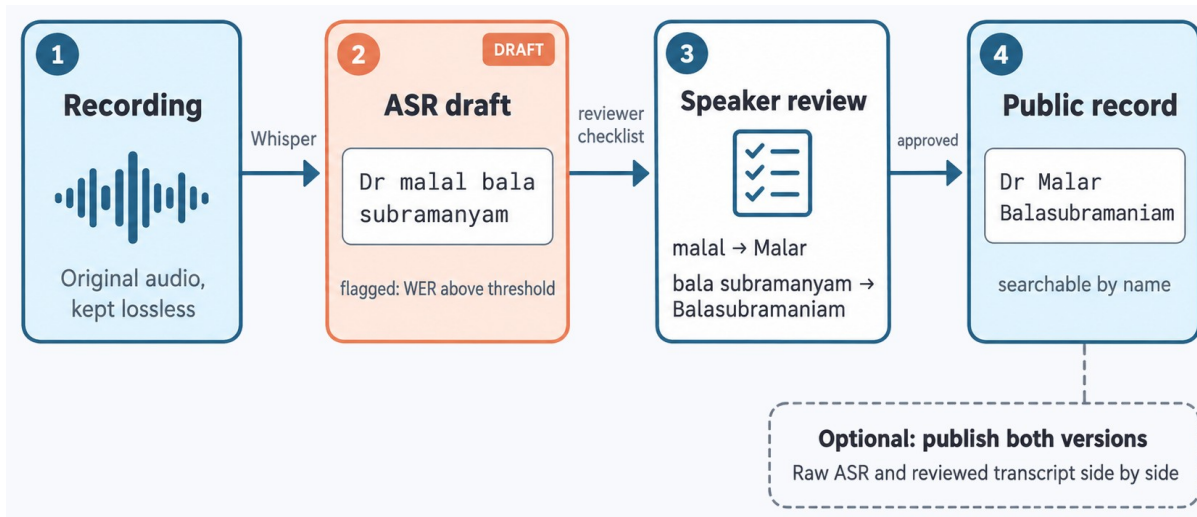
No marker distinguishes translated from hallucinated spans, although the task token specified transcription, not translation.



ASR sits at Trouillot's seam, so speaker review must come first

Trouillot's four moments of historical production [11]

- 1 Fact creation**
making sources
- 2 Fact assembly**
making archives
----- ASR sits here
- 3 Fact retrieval**
making narratives
- 4 Retrospective significance**
making history



Draft flag

Above a WER threshold, mark the transcript draft until a speaker reviews it

Checklist

Review with the error typology, not a general 'check the transcript'

Transparency

Publish raw and reviewed transcripts side by side [12]



Conclusions & Future Work

- **Structural gap:** South Asian WER is 3.8–11× English on read speech; all four Benjamini–Hochberg $q \leq 0.001$
- **Fine-tuning helps, not enough:** –54 to –67% WER; Bengali still ~5× English
- **Errors differ in consequence:** dropped and anglicised words vanish from search; WER alone cannot show which
- **Silent failure:** a forced-English hint gives fluent translation plus hallucination at a reported LID of 1.0
- **For archives:** treat transcripts as drafts; review against the typology before indexing
- **Future work:** language-hint robustness, a translate-vs-transcribe signal, spontaneous code-switched training data
- **Limits:** read speech (a best case), one case-study clip, Hindi/Tamil fine-tunes saw FLEURS, Evaluate one model family



Code & results

Thank You

Questions and feedback welcome

github.com/sayanmndl/whisper-archival-bias

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Code & results

Backup slides

Protocol · normaliser · examples · limitations · references · Whisper architecture



Protocol: read speech, bootstrap confidence intervals, corrected tests

Language	FLEURS config	Clips
English (US)	en_us	600
Hindi	hi_in	418
Bengali	bn_in	600
Tamil	ta_in	584
Urdu	ur_pk	299

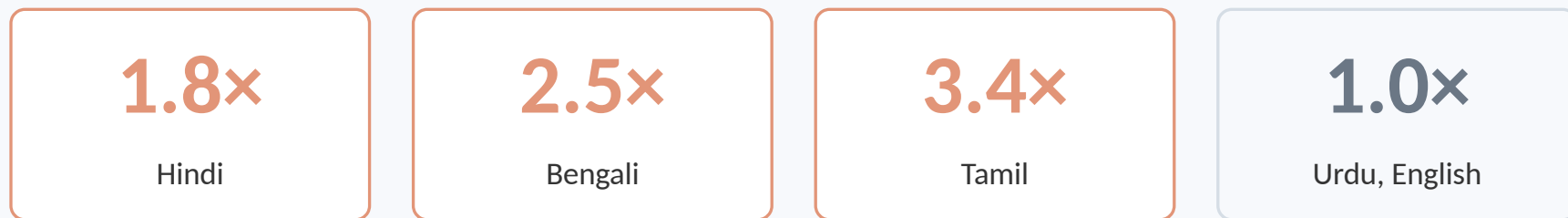
- **WER** = (substitutions + deletions + insertions) / words in the reference
- **95% bootstrap CI** (1000 resamples, seed 13)
- **Unpaired bootstrap test** vs. English, Benjamini-Hochberg corrected over the 4 comparisons
- **Hardware:** one RTX 4070 Ti SUPER (16 GB)

Scope: FLEURS is read studio speech, not spontaneous archive audio.

The reported WERs should be read as a lower bound for diasporic recordings, which are spontaneous, accented and often code-switched. How much worse those recordings fare is not measured here.



The standard normaliser splits Indic words into consonant fragments



Increase in reference token count under Whisper's BasicTextNormalizer

One word, two normalisers: कार्यक्रमों (*karyakramon*, “programmes”)

Whisper-Radford	क र य क र म	1 word → 4 fragments: vowel signs (matras) become spaces. We use it only to compare with published WER.
Ours	कार्यक्रमों	1 word stays 1 word. We use it wherever word identity matters, e.g. the typology.

Ours = Unicode NFC + lowercase + strip punctuation. NFC (Normalization Form C) stores characters that look the same with the same bytes. Tamil கௌ can be typed as one code point or two; after NFC both match, so they aren't scored as an error.



Dropped and anglicised spans erase terms from the index

Language	Error	Reference	Sounds like • means	Whisper output
Hindi	dropped	10-60 मिनट	<i>das-saath minat</i> 10–60 minutes	<dropped>
Bengali	dropped	বিশ্ববিদ্যালয়ের গবেষকরা	<i>bishshobiddalayer gobeshokra</i> university researchers	<dropped>
Tamil	dropped	செயல்படவில்லை	<i>seyalpadavillai</i> does not work	<dropped>
Hindi	anglicised	यूनिवर्सिटी ऑफ़ डंडी	<i>yunivarsiti of dandi</i> University of Dundee	university of dundee
Bengali	anglicised	অ্যারোস স্মিথ	<i>aerosmith</i> Aerosmith (the band)	aerosmith
Tamil	anglicised	ஸ்காப்டிங் போல்டு	<i>skaffoldu</i> scaffold	skaf bold

A wrong word is still searchable. A dropped or anglicised one is gone for anyone searching in the original script.



Limitations: read speech, one clip, fine-tunes that saw FLEURS

Register

FLEURS is read studio speech; the case study shows a failure mode exists, not how common it is.

→ **Treat the numbers as a best case.**

Fine-tune comparison

No comparable Urdu fine-tune. The Hindi and Tamil fine-tunes trained on FLEURS train/dev.

→ **Their gains are optimistic.**

Typology

Automatic, word-by-word tagging shows a reviewer where to look.

→ **It does not replace review by a speaker of the language.**

Scope and access

No commercial ASR APIs. SAADA and the 1947 Partition Archive require written permission.

→ **Their audio needs their research-access protocols.**



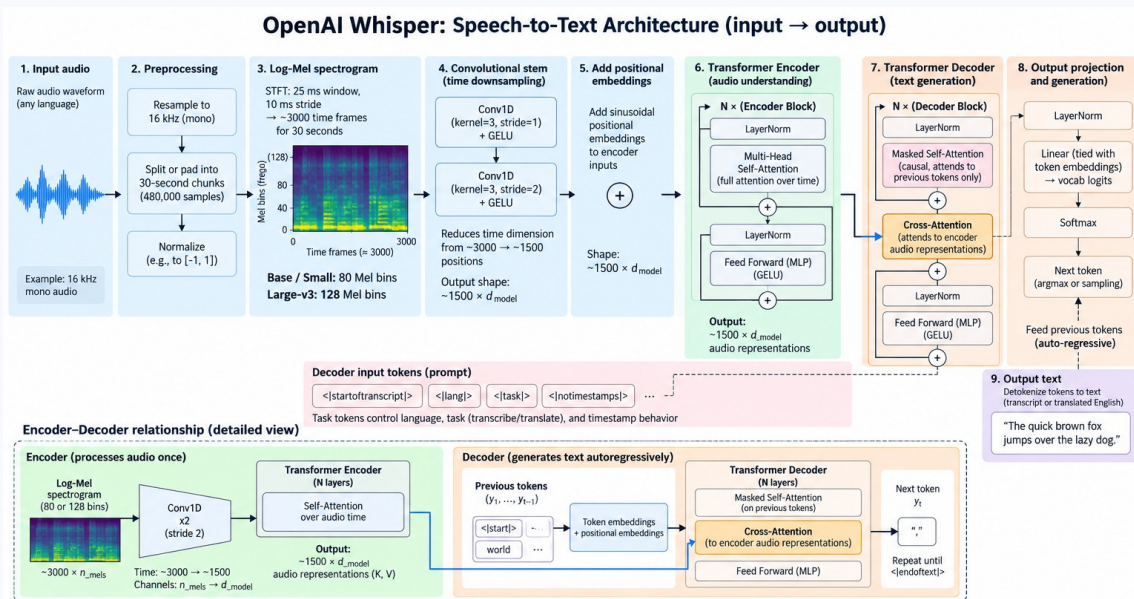
References

Numbered for this talk; full bibliography in the paper.

- [1] A. Portelli, “What makes oral history different,” in *The Death of Luigi Trastulli and Other Stories*, SUNY Press, 1991.
- [2] J. Sproule, “The ground truth gradient: When AI-assisted transcription becomes historical record,” *On History (IHR blog)*, 2025.
- [3] A. Radford et al., “Robust speech recognition via large-scale weak supervision,” *ICML*, 2023.
- [4] A. Biswas et al., “Adapting Whisper for low-resource Hindi–English code-mix speech...,” *Interspeech*, 2025.
- [5] P. Jain and A. Bhowmick, “Analyzing code-switching scenarios in India’s diverse linguistic landscape...,” *Comput. Electr. Eng.*, 2025.
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- [7] A. Conneau et al., “FLEURS: Few-shot learning evaluation of universal representations of speech,” *IEEE SLT*, 2023.
- [8] K. S. Bhogale et al., “Vistaar: Diverse benchmarks and training sets for Indian language ASR,” *Interspeech*, 2023.
- [9] Hugging Face model cards: [vasista22/whisper-hindi-large-v2](#), [anuragshas/whisper-large-v2-bn](#), [vasista22/whisper-tamil-large-v2](#).
- [10] Anonymous narrator, anecdotal Bengali recording of a Partition-era recollection, 98 s, consent on file with the authors, 2026.
- [11] M.-R. Trouillot, *Silencing the Past: Power and the Production of History*, Beacon Press, 1995.
- [12] M. Frisch, *A Shared Authority: Essays on the Craft and Meaning of Oral and Public History*, SUNY Press, 1990.
- [13] T. Javed et al., “IndicVoices: Towards building an inclusive multilingual speech dataset for Indian languages,” *Findings of ACL*, 2024.



Appendix A: Whisper architecture and the paper's failure modes



Model size comparison (Whisper)

Model	Parameters (approx)	Encoder layers (N)	Decoder layers (N)	Hidden size (d _{model})	Attention heads (n _{heads})	Mel bins (n _{mels})	Audio frames (for 30 s)	Notes
Base	~74M	6	6	512	8	80	~3000 → ~1500	Good balance of speed and accuracy.
Small	~244M	12	12	768	12	80	~3000 → ~1500	Higher accuracy, widely used.
Large-v3	~1.55B	32	32	1280	20	128	~3000 → ~1500	Highest accuracy, multilingual, better robustness.

- Same basic architecture for all sizes; scale changes mainly depth (number of layers), width (hidden size), attention heads, and parameter count.
- In the current OpenAI Whisper repository, the model alias "large" points to large-v3. Earlier large-v1/v2 used 80 Mel bins (instead of 128).

<|lang|> token (box 'Decoder input tokens')

Predicted by the model, or set by the caller. Forcing it to English is the case-study experiment.

One language per 30 s chunk (box 2)

The model commits to a single language per window, so code-switched speech is flattened to one language.

Decoder = language model on audio (box 7)

When acoustic evidence is weak or mismatched, the text prior takes over and writes fluent output: the hallucination mechanism.

